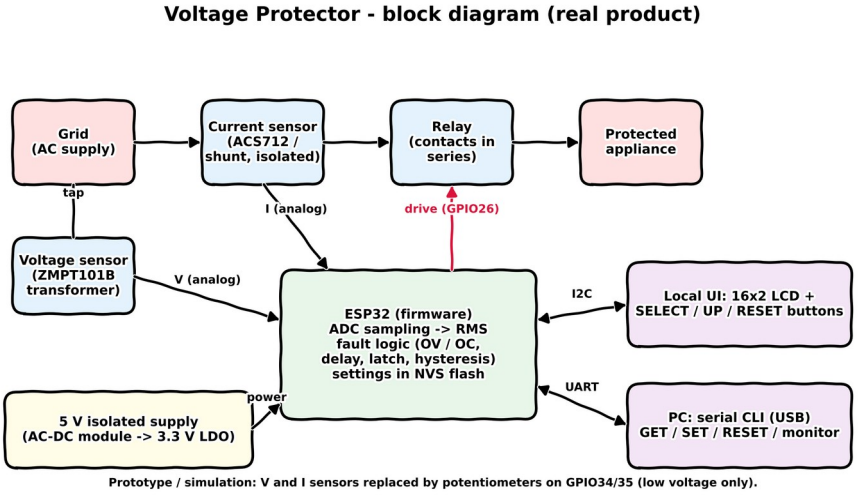
Voltage Protector - Demonstration Report (DRAFT: rewrite in your own words)

Design approach

An ESP32 reads a voltage channel (ZMPT101B in the real product) and a current channel (ACS712) through its 12-bit ADC, computes RMS values, and opens a relay in series with the appliance when either exceeds a user-set limit. The architecture follows the ESP32 sensing / relay / LCD design from the course project deck. I separated the decision logic (protector_core.h, no hardware calls) from the hardware layer so the exact code that runs on the device can be tested on a PC.

Implementation

Protection: over-voltage default 250 V, over-current 10 A. A fault must persist for a settable delay (default 20 ms) so short glitches do not cause nuisance trips, but readings above 115% (V) or 150% (I) of the limit trip on the first sample. After a trip the relay stays latched open; reset is refused until readings are below 97% of the limit (hysteresis). Optional auto-recover re-closes after readings stay safe for a set time. **Settings:** from the device with SELECT/UP/RESET buttons and a 16x2 LCD, or from a computer over USB serial (GET, SET OV|OC|DELAY|AUTO|RECOVER, RESET) using a small Python CLI; values are stored in flash. **Safety:** nothing is built on mains; potentiometers stand in for the sensor outputs (0-3.3 V = 0-300 V / 0-20 A).



Results

Test (host run, same core code)	Result
265 V sustained, 20 ms delay	Tripped, latency 20 ms (= set delay)
10 ms spike to 265 V	Ignored, no trip
300 V surge / 18 A short circuit	Tripped on first sample (0 ms)
12 A overload	Tripped, fault = OC
Reset while fault present / after it clears	Refused / accepted
SET OV 999 (out of range), unknown command	Rejected with ERR, setting unchanged

What is not complete (and why)

21/21 host checks pass, but they test the logic only. The firmware has **not** been compiled or run on a physical ESP32 and the Wokwi project has not been simulated by me; the AC RMS sampling path and sensor calibration are written but untested; the 20 ms is a configured delay, not a measured relay-opening time (the relay itself adds roughly 10 ms). Next steps: run it in Wokwi, capture serial logs and a scope trace of the relay output, and calibrate the sensors against a reference meter.